Musterwerk Automotive GmbH

Lastenheft: Autonomous Emergency Braking (AEB)

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Revision History

Version	Date	Author	Change
0.1	2026-10-27	B. Aksakal	Initial draft
0.2	2026-10-27	B. Aksakal	Added system status, failure handling and diagnostics

1 Introduction

This document specifies the requirements for the Autonomous Emergency Braking (AEB) system of passenger cars (category M1). It is intended for suppliers responding to the request for quotation. The supplier shall confirm each requirement in the compliance matrix delivered with the quotation.

2 Scope

The AEB system detects vehicles ahead and warns the driver. If the driver does not react, the system brakes automatically to avoid or mitigate a collision. Pedestrian and cyclist detection are out of scope for this version of the document and will be specified separately.

3 Activation and Braking

This section describes when the system is active and how the emergency braking phase is performed.

AEB-001 Operating speed range

The AEB system shall be active at vehicle speeds between 10 km/h and 60 km/h.

AEB-002 Minimum activation speed

The AEB system shall not be active below 15 km/h.

AEB-003 Braking reaction

The system shall brake fast when a collision is imminent.

AEB-004 Speed reduction

In the stationary target test scenario, the emergency braking phase shall reduce the vehicle speed by at least 20 km/h before the estimated point of impact, measured from the start of the emergency braking phase.

AEB-005 Earliest braking start

The emergency braking phase shall not start earlier than 3.0 s before the estimated time to collision.

AEB-006 Weather robustness

The system shall work reliably in all weather conditions.

Note: Speed values refer to the speed shown on the vehicle speedometer.

4 Driver Warning and Interaction

AEB-007 Collision warning

The system shall warn the driver at least 0.8 s before emergency braking starts.

AEB-008 Warning type

The collision warning shall be acoustic and visual.

AEB-009 Driver override

The driver shall be able to interrupt emergency braking by pressing the accelerator pedal.

AEB-010 Warning perceptibility

The collision warning shall be clearly perceptible to the driver.

5 System Status and Deactivation

The driver may deactivate the system manually, for example during off-road driving or when towing. The following requirements limit how deactivation is done and how the system status is shown.

AEB-011 Automatic reactivation

The system shall be automatically reactivated at the start of each engine start cycle.

AEB-012 Manual deactivation

Manual deactivation of the system shall require at least two deliberate actions by the driver, such as selecting a menu entry and confirming the selection, so that the system cannot be switched off by a single accidental button press.

AEB-013 Deactivation indication

When the system is deactivated, a constant optical warning signal shall be shown to the driver.

AEB-014 Failure warning time

A failure warning signal shall be displayed within 10 s after a system failure is detected.

6 Diagnostics and Robustness

AEB-015 Diagnostic trouble codes

The system shall store a diagnostic trouble code for each detected sensor fault.

AEB-016 False interventions

The system shall avoid false braking interventions as far as possible.

AEB-017 Sensor range

The radar sensor shall detect vehicles ahead at a distance of at least 150 m.

AEB-018 Failure warning time (diagnostics)

The failure warning signal shall be displayed within 5 s after a system failure is detected.

Note: Diagnostic trouble codes follow the format defined in the Musterwerk diagnostics guideline.